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carbon bearing positive charge, greater is number of the hyper conjugate structure. thus the stability of various carbocations decreases in the order

3^0 Carbocation > 2^0 Carbocation > 1^0 Carbocation

12.2 Different types of organic reactions

Organic compounds undergo many number of reactions, however in actual sense we can fit all those reactions into the below mentioned six categories.

- Substitution reactions
- Addition reactions
- Elimination reactions
- Oxidation and reduction reactions
- Rearrangement reactions
- Combination of the above

12.2.1 Substitution reaction (Displacement reaction)

In this reaction an atom or a group of atoms attached to a carbon atom is replaced by a new atom or a group of atoms. Based on the nature of the attacking reagent, this reactions can be classified as

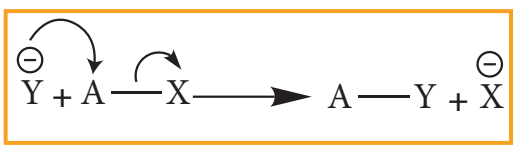
i) Nucleophilic substitution

ii) Electophilic substitution

iii) Free radical substitution

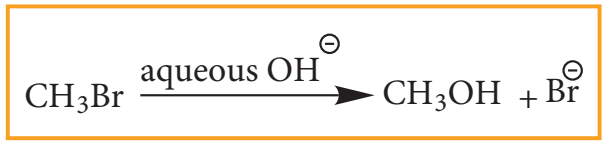
Nucelophilic substituion:

This reaction can be represented as



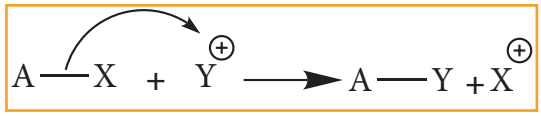
Here Y^- is the incoming nucleophile or and attacking species and x^- is the leaving group.

Example: Hydrolysis of alkyl halides



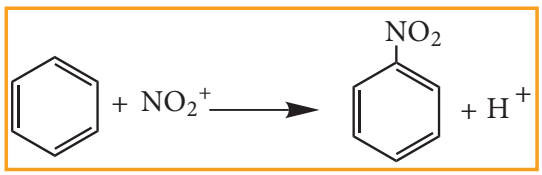
Aliphatic nucleophilic substitution reactions take places either by S_N1 or S_N2 mechanism. Detailed study of the mechanisms is given in unit 14.

Electrophilic Substitution



Here Y^+ is an electrophile

Example: Nitration of Benzene



Mechanism of aromatic electrophilic substitution reactions (EAS) is discussed in detail in unit 13.

Free radical substitution





Aliphatic electrophilic substitution

A general aliphatic electrophilic substitution is represented as



